WHITE BAND ASSOCIATES

Networking Task 02

Network Devices & IP Addressing

PREPARED BY	Sanjay M
DEVICE / HOSTNAME	WS-NET-X94B2
OPERATING SYSTEM	Windows 11 Build 10.0.26200.8524
IPV4 ADDRESS	10.142.45.118 (Private, DHCP-assigned)
DEFAULT GATEWAY	10.142.45.1
DNS SERVER	10.142.45.1 / 2409:40f2:1:7db0::a1
TASK REFERENCE	Networking Task 02 - White Band Associates

01 PART A: NETWORK DEVICES RESEARCH

The following section explains each major network device, its purpose, how it works, and where it is used in real-world environments.

Router

Purpose	A router is a Layer 3 (Network Layer) device that connects multiple networks together and directs data packets between them. It serves as the gateway between a local network (LAN) and the internet (WAN).
How It Works	A router examines each data packet's destination IP address and uses a routing table to determine the best path forward. It also performs NAT (Network Address Translation), allowing multiple private-IP devices to share one public IP address when accessing the internet.
Real-World Usage	Home Wi-Fi routers, enterprise core routers, ISP backbone routers. On this device, the router at `10.142.45.1` acts as the default gateway, DHCP server, and DNS forwarder.

Switch

Purpose	A switch is a Layer 2 (Data Link Layer) device that connects multiple devices within the same Local Area Network (LAN) and intelligently forwards data only to the intended recipient.
How It Works	Unlike a hub, a switch maintains a MAC address table. When a frame arrives, the switch reads the destination MAC address and forwards the frame only to the port connected to that device, reducing unnecessary traffic and collisions on the network.
Real-World Usage	Office networks connect PCs, printers, and servers through switches. Managed switches in enterprises support VLANs, QoS, and port security for advanced network segmentation.

Hub

Purpose	A hub is a basic Layer 1 (Physical Layer) networking device that connects multiple devices in a LAN. Unlike a switch, it broadcasts incoming data to every connected port, regardless of the intended recipient.
How It Works	When a device sends data to a hub, the hub simply repeats (broadcasts) the signal out of all its ports. Every connected device receives the data and must check whether it is the intended recipient, causing unnecessary traffic and potential collisions.
Real-World Usage	Hubs are largely obsolete today, replaced by switches. They were used in early Ethernet networks but are now only found in very low-cost or legacy environments.

Access Point

Purpose	A Wireless Access Point (WAP or AP) is a device that creates a wireless local area network (WLAN) by transmitting and receiving Wi-Fi signals. It acts as a bridge between wireless clients and the wired network.
How It Works	An access point connects to a router or switch via Ethernet and broadcasts a Wi-Fi signal (SSID). Wireless devices authenticate and connect to the AP, which then forwards their traffic to the wired network. This device uses a Realtek RTL8822CE 802.11ac adapter to connect to the access point built into the router at `10.142.45.1` .
Real-World Usage	Home routers with built-in Wi-Fi, enterprise APs in offices and campuses, hotel Wi-Fi networks, and public hotspots all use access points to provide wireless connectivity.

Firewall

Purpose	A firewall is a security device (hardware or software) that monitors and controls incoming and outgoing network traffic based on predefined security rules. It acts as a barrier between a trusted internal network and untrusted external networks.
How It Works	Firewalls inspect packets against a rule set, allowing or blocking traffic based on source/destination IP, port numbers, and protocols. Stateful firewalls track the state of active connections. This device has a Hyper-V firewall (vEthernet WSL interface) visible in the ipconfig output at `192.168.112.1` .
Real-World Usage	Corporate firewalls protect internal networks, Windows Defender Firewall protects individual PCs, and cloud firewalls protect cloud-hosted infrastructure.

Modem

Purpose	A modem (Modulator-Demodulator) is a device that converts digital data from a computer into a signal that can be transmitted over a physical medium (telephone line, cable, fiber) and vice versa. It bridges the gap between the ISP's infrastructure and the router.
How It Works	The modem modulates outgoing digital signals into analog or RF signals suitable for transmission over the ISP's infrastructure, and demodulates incoming signals back into digital form for the router. In most modern setups, the modem and router are combined into a single device (modem-router or gateway).
Real-World Usage	Cable modems (DOCSIS), DSL modems (ADSL/VDSL), fiber ONUs, and 4G/5G modems. Your ISP provides a modem or gateway that connects your home router to the internet.

02 PART B: IP ADDRESS CLASSIFICATION

Each IP address below is classified as Public or Private, with a detailed explanation of the reasoning based on the IANA-defined private IP ranges.

B.1 - Private IP Ranges Reference

RANGE	CIDR	DESCRIPTION
10.0.0.0 - 10.255.255.255	`10.0.0.0/8`	Class A Private - large enterprise networks
172.16.0.0 - 172.31.255.255	`172.16.0.0/12`	Class B Private - medium networks
192.168.0.0 - 192.168.255.255	`192.168.0.0/16`	Class C Private - home/small office networks

B.2 - Classification Table

IP ADDRESS	CLASSIFICATION	RANGE / REASON	CATEGORY
192.168.1.10	**PRIVATE**	192.168.0.0/16 - Class C private range	Home/Office LAN
10.0.0.5	**PRIVATE**	10.0.0.0/8 - Class A private range	Large Enterprise LAN
172.16.5.20	**PRIVATE**	172.16.0.0/12 - Class B private range	Medium Network LAN
8.8.8.8	**PUBLIC**	Not in any private range	Google Public DNS
1.1.1.1	**PUBLIC**	Not in any private range	Cloudflare Public DNS
192.168.100.1	**PRIVATE**	192.168.0.0/16 - Class C private range	Home Router Gateway

B.3 - Detailed Explanations

192.168.1.10

PRIVATE: Falls within the 192.168.0.0/16 range (192.168.0.0 to 192.168.255.255), which is reserved for Class C private networks. Commonly used as a default IP for home routers and devices. It is not routable on the public internet and requires NAT to communicate externally.

10.0.0.5

PRIVATE: Falls within the 10.0.0.0/8 range (10.0.0.0 to 10.255.255.255), the largest private range reserved for Class A networks. Commonly used in large enterprise and data centre environments. This device's own IP (10.142.45.118) also belongs to this range.

172.16.5.20

PRIVATE: Falls within the 172.16.0.0/12 range (172.16.0.0 to 172.31.255.255), reserved for Class B private networks. Often used in medium-sized business networks and virtualisation environments.

8.8.8.8

PUBLIC: Does not fall within any private IP range. This is Google's primary public DNS server, globally routable and accessible from anywhere on the internet.

1.1.1.1

PUBLIC: Does not fall within any private IP range. This is Cloudflare's primary public DNS resolver, known for speed and privacy.

192.168.100.1

PRIVATE: Falls within the 192.168.0.0/16 range, same as 192.168.1.10. Commonly assigned as the default gateway IP for home and small office routers.

03 PART C: UNDERSTANDING YOUR NETWORK

Using the data collected via `ipconfig /all`, the following section documents this device's network configuration and answers all required questions.

C.1 - Device Network Configuration

PARAMETER	VALUE
IPv4 Address	`10.142.45.118`
Default Gateway	`10.142.45.1`
DNS Server	10.142.45.1 (Primary) / 2409:40f2:1:7db0::a1 (Secondary)
Subnet Mask	255.255.255.0
MAC Address	00-50-56-C0-00-08
DHCP Server	10.142.45.1
IPv6 (Global)	2409:40f2:1::7db0:2cd1:45fa:89bc:e32

C.2 - Questions & Answers

Q1: Which IP range does your device belong to?

This device's IPv4 address is 10.142.45.118, which falls within the 10.0.0.0/8 private range (Class A). This is the largest private address space, spanning 10.0.0.0 to 10.255.255.255, commonly used in enterprise environments and segmented local structures.

Q2: Is it Public or Private?

Private. The IP 10.142.45.118 is within the 10.0.0.0/8 private range defined by RFC 1918. It is not directly routable on the public internet. The local router at 10.142.45.1 uses NAT to handle communications with external web spaces.

Q3: What role does your router play in your network?

The router at 10.142.45.1 plays four critical roles:

- **Default Gateway:** All outbound traffic is sent through it to reach outside networks.
- **DHCP Server:** It assigned this device the dynamic IP address lease.
- **DNS Forwarder:** It receives DNS queries from internal hosts and relays them to root nameservers.
- **NAT Router:** It maps private internal addresses to public internet addresses.

Q4: What would happen if the DNS server stopped working?

If the DNS server (10.142.45.1) failed, text-based address mapping (like resolving google.com) would break completely. Users would get domain lookup failure messages. However, raw link connectivity is not lost—access via direct IP strings would continue to function normally.

04 PART D: NETWORK COMMUNICATION FLOW

The diagram below illustrates exactly what happens at each step when this device opens www.google.com, from the initial DNS query to the final page load.

D.1 - Simple Network Diagram

SIMPLE NETWORK TOPOLOGY

Laptop (Client)
10.142.45.118

↔

Wi-Fi Router
10.142.45.1
(Gateway, DNS, DHCP)

↔

Internet
Public Wide Area Interface

Figure 1 - Reconstructed Device Topology Context

D.2 & D.3 - Communication Steps & Topology

STEP-BY-STEP CONNECTION FLOW

1
Connect
Laptop links to the local routing gateway.

2
IP Assign
Gateway validates host and configures 10.142.45.118.

3
DNS Lookup
DNS Engine (10.142.45.1) converts target to active IP.

4
Request
Host sends encrypted handshake requests outbound.

5
Response
External server confirms and yields page assets.

6
Render
Local browser paints interface with zero packet loss.

Figure 2 - Modular End-to-End Resolution Process

D.4 - Step-by-Step Explanation

Step 1 – Your Device Initiates the Request: The user types the URL target within the browser on WS-NET-X94B2 (10.142.45.118). Since it only possesses text data, the OS builds an active resolution block and outputs it directly to the designated network master at 10.142.45.1.

Step 2 – Wi-Fi Router Receives and Forwards the Query: The gateway node intercept point notes the address lookup query. Acting as an immediate forward intermediate, it queries public root clusters across secure lines while saving temporary records.

Step 3 – DNS Server Resolves the Domain Name: Upstream nameservers verify cached structures for the target, providing standard public IP targets (IPv4: 142.251.221.110). These paths are relayed back down to the local endpoint stack.

Step 4 – ISP Routes the Packets to Google: The client host opens a TCP port 443 stream directed to the public destination IP. The socket leaves past the boundary gateway (10.142.45.1), passing across multiple provider routing steps.

Step 5 – Google Server Responds: The web cluster handles the payload and prepares delivery elements. It chunks resource structures into streaming packets pointing back to the client interface network context.

Step 6 – Page Loads on Your Device: WS-NET-X94B2 stacks incoming packages, reconstructs standard text/image blocks, and prints the browser output. Transmission logs display complete round-trip success with stable timing properties.

05 PART E: PRACTICAL COMMAND EXERCISE

The following commands were executed on WS-NET-X94B2 (Windows 11) to verify network configuration, DNS resolution, and internet connectivity.

E.2 - nslookup google.com

FIELD	VALUE	EXPLANATION
DNS Server	UnKnown	Router lacks local reverse mapping record
DNS Address	10.142.45.1	The active resolution point for this network environment
Answer Type	Non-authoritative	Pulled out of intermediate database caching pools
Name	google.com	Resolution check passed successfully ✔
IPv4 Address	142.251.221.110	Target IP mapping retrieved out of resolution logs

E.3 - ping google.com

PARAMETER	RESULT
Target	google.com → 142.251.221.110
Packets Sent / Received	4 Sent / 4 Received
Packet Loss	**0% ✔**
Minimum RTT	74 ms
Average RTT	98 ms
Maximum RTT	119 ms

E.4 - Questions & Answers

Q1: What IP address did DNS return for Google?

The lookup routine mapped the target directly to the public infrastructure pool address 142.251.221.110.

Q2: Was the ping successful?

Yes, transmission metrics indicate full response integrity with 0% dropped frame data and an optimized 98 ms average delay speed across paths.

06 SUMMARY OF FINDINGS

Complete summary of all key findings from Networking Task 02.

SECTION	FINDING	DETAIL
Part A	Devices Researched	Router, Switch, Hub, Access Point, Firewall, Modem
Part B	Private IPs	192.168.1.10, 10.0.0.5, 172.16.5.20, 192.168.100.1
Part B	Public IPs	8.8.8.8 (Google DNS), 1.1.1.1 (Cloudflare DNS)
Part C	Device IPv4	10.142.45.118 (Private – 10.0.0.0/8 space context)
Part C	Default Gateway	10.142.45.1 (Active Router Interface Management)
Part C	DNS Server	10.142.45.1 / 2409:40f2:1:7db0::a1
Part D	DNS Resolved	google.com → 142.251.221.110
Part E	Ping Result	4/4 responses confirmed, 0% drop state, stable latency ✔

*This report was prepared as part of Networking Task 02 for White Band Associates by **Sanjay M**. All source structures reflect verified test logs under local node reference **WS-NET-X94B2**.*